

NOTHINGDESKTOP

Minimalist Dot-Matrix PC Widget Suite for Windows & macOS

Project Title:	NothingDesktop – Cross-Platform PC Widget Suite
Domain / Stack:	Advanced Java (Spring Boot 3, SQLite, WebSockets, OSHI, HTML5/CSS3)
Degree & Year:	BSc IT – 3rd Year (Final Year Project Submission)
Target OS:	Windows 10 / 11 & macOS

1. Executive Summary & Problem Statement

Modern operating systems like Windows and macOS provide rigid, locked-down widget environments that are often confined to taskbars or hidden sidebars. Existing third-party desktop customization platforms (like Rainmeter) are 15-year-old legacy Windows-only applications that require users to manually write complex .ini scripts.

NothingDesktop brings the iconic minimalist **Nothing OS dot-matrix aesthetic** to desktop computers. It provides users with a modular, interactive desktop widget suite featuring a visual drag-and-drop customizer, real-time hardware monitoring, quick app launchers, and an innovative **zero-code REST API widget builder** that allows users to turn any web API into a dot-matrix widget.

/// NOTHINGDESKTOP CONCEPTUAL UI HUD ///	
■ Dot Clock: 10:42:05 PM	■ Hardware: CPU: 38% RAM: 6.2 GB
■ Launcher: [VS Code] [Terminal] [Chrome]	■ Live API: BTC: \$64,200 Weather: 24°C

2. Key System Modules

1. Dot-Matrix Clock & Timer Digital/Analog Nothing-style clocks, timezone indicators, and Pomodoro study timer.	2. Hardware System Monitor Real-time CPU %, RAM usage, battery level, and disk space telemetry using Java OSHI API.
3. App & Script Launcher Launch desktop executables (code.exe, cmd.exe) with 1-click dot-matrix buttons.	4. Dynamic REST API Engine Paste any REST API URL + key (Weather, Crypto, GitHub) to auto-render a live widget.
5. Quick Task & Note Board Minimalist dot-matrix sticky note board and task checklist for daily productivity.	6. Drag & Drop Visual Grid Interactive grid customizer to move, resize, and lock dot-matrix widgets visually.

3. Technical Architecture & Advanced Java Stack

Layer	Technology Used	Role in Project
Backend Core	Java 21 + Spring Boot 3.x	Local server managing widget state, API polling, and system tray lifecycle.
Hardware Telemetry	OSHI (Operating System & Hardware Info)	Native Java hardware library reading CPU, Memory, Disk, and Battery stats.
App Execution	Java ProcessBuilder API	Executes local PC binaries and shell scripts safely on user click.
Database	SQLite / H2 + Spring Data JPA	Embedded database storing widget X/Y positions, dimensions, and user settings.
Real-Time Gateway	Spring WebSockets (STOMP/SockJS)	Sub-second real-time streaming of clock ticks, CPU %, and live API feeds.
Frontend UI	HTML5, CSS3 (Nothing Theme), JavaScript	Dot-matrix styling, monochrome dark mode, grid engine, and WebSockets.

4. Innovation & Competitive Analysis

Unlike standard CRUD projects, **NothingDesktop** targets real-world daily utility and showcases deep Java system programming capabilities.

Feature	Rainmeter (Legacy)	Windows/Mac Widgets	■ NothingDesktop
OS Support	Windows Only	Locked Ecosystem	Cross-Platform (Win & Mac)
Configuration	Editing .ini code	Fixed widgets only	Visual Drag & Drop GUI
Custom Web APIs	Requires Lua scripting	Not supported	Zero-Code REST API Parser
Tech Stack	C++ / Lua (2005)	Proprietary OS APIs	Java 21 & Spring Boot

5. Project Implementation Timeline

- **Phase 1 (Week 1):** Architecture finalization, Spring Boot initialization, SQLite schema setup.
- **Phase 2 (Week 2):** OSHI hardware reader integration, Java ProcessBuilder app execution engine.
- **Phase 3 (Week 3):** WebSocket real-time gateway implementation, dynamic REST API parsing service.
- **Phase 4 (Week 4):** Nothing OS dot-matrix UI frontend development, CSS grid layout builder.
- **Phase 5 (Week 5):** Integration testing, cross-platform validation (Windows/Mac), project documentation.